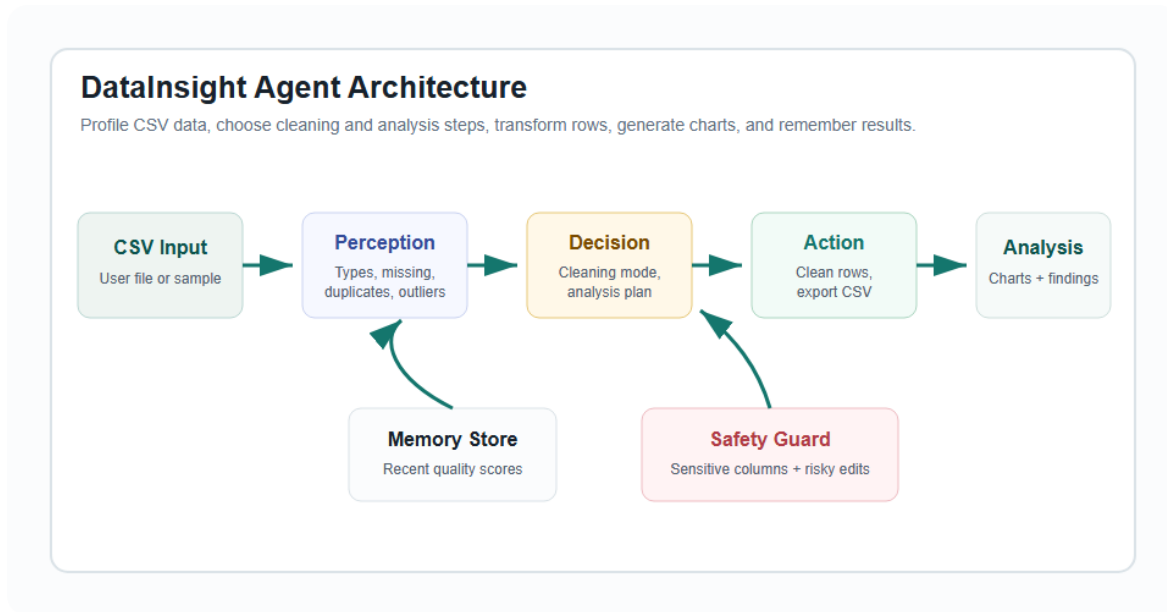


DataInsight Agent

COMPSCI 767 Assignment 2 - Intelligent CSV Cleaning and Analysis Agent

GitHub repository: <https://github.com/dsfga/datainsight-agent>

System Design



Design Explanation

- Perception parses CSV rows and profiles column types, missing values, duplicate rows, category variation, outliers, and possible sensitive columns.
- Decision logic chooses a cleaning plan from conservative, balanced, or aggressive mode and selects analysis steps from available numeric, categorical, and date fields.
- Action transforms the dataset, exports cleaned CSV, renders charts, summarizes findings, and records a lightweight memory checkpoint.
- Safety mechanisms keep processing local, flag possible sensitive columns, and raise risk when cleaning may change numeric meaning.

How the Prototype Works

Screenshots from the running local web app

CSV CLEANING AND ANALYSIS AGENT

DataInsight Agent

Perceives data quality issues, decides cleaning and analysis steps, then acts by transforming the dataset, plotting charts, and summarizing findings.

Local prototype. Use sample or non-sensitive CSV data only.

选择文件 从数据库中提取文件

Cleaning mode

Balanced

Analysis goal

Sales analysis

CSV preview / paste area

data,region,product,units,revenue,customer rating

2026-01-02,north,alpha,12,2400,4.8

2026-01-05,north,alpha,14,2800,4.7

2026-01-08,south,beta,7,1050,4.1

2026-01-12,east,gamma,5,1800,3.8

2026-01-18,west,alpha,20,4000,4.9

2026-02-01,north,beta,11,1650,4.2

2026-02-03,south,gamma,15,2250,4.5

2026-02-05,north,alpha,10,2000,4.6

Load Sample

Run Agent

Cleaned CSV

Clear Memory

Demo view: balanced CSV cleaning and analysis

Perceive

Decide

Clean

Analyze

Remember

Rows

24

Columns

6

Before quality

94/100

After quality

98/100

AGENT DECISION

Balanced cleaning + 4 analysis steps

Mode balanced selected 7 cleaning action(s) for 24 row(s). The agent prioritized missing values, duplicates, category consistency, and outlier handling. 4 analysis step(s) were selected from detected numeric, categorical, and date columns.

Low Risk

PERCEPTION

Data profile

COLUMN	TYPE	MISSING	OUT
Date	date	0	0
Region	category	1	0
Product	category	0	0
Units	number	1	0
Revenue	number	1	1
Customer Rating	number	0	0

DECISION

Cleaning strategy

1. remove

duplicate rows - 1 duplicate row(s) detected.

2. fill unknown

Region - 1 missing value(s) in Region.

3. standardize category

Region - Normalizer capitalization and spacing in Region.

4. standardize category

Product - Normalizer capitalization and spacing in Product.

5. fill median

Units - 1 missing value(s) in Units.

6. fill median

Revenue - 1 missing value(s) in Revenue.

7. flag outliers

Revenue - 1 numeric outlier(s) detected in Revenue.

ACTION

Cleaning actions applied

Removed 1 duplicate row(s).

Filled missing Region values with Unknown.

Standardized category labels in Region.

Standardized category labels in Product.

Filled missing Units values with 11.

Filled missing Revenue values with 1800.

Flagged outliers in Revenue for review.

Quality (M): +4

ANALYSIS

Generated charts

ANALYSIS

Generated charts

Units by Region

Units trend by month

Units vs Revenue

SUMMARY

Findings

Data quality improved from 94/100 to 98/100 after cleaning.

North has the highest Units total in the category breakdown.

Units increased from 2026-01 to 2026-05.

PREVIEW

Cleaned data sample

DATE	REGION	PRODUCT	UNITS	REVENUE	CUSTOMER RATING
2026-01-02	North	Alpha	12	2400	4.8
2026-01-05	North	Alpha	14	2800	4.7
2026-01-08	South	Beta	7	1050	4.1
2026-01-12	East	Gamma	5	1800	3.8
2026-01-18	West	Alpha	20	4000	4.9
2026-02-01	North	Beta	11	1650	4.2

Balanced mode: the agent perceives data quality problems and chooses seven low-risk cleaning actions.

Results view: the agent acts by plotting charts, previewing cleaned data, and summarizing findings.

Workflow Evidence

- Run Agent produces a visible perceive-decide-act pipeline from the sample CSV.
- Balanced mode flags outliers for review; aggressive mode caps them and marks medium risk.
- The chart panel shows the agent selected analysis actions based on detected columns.
- The cleaned CSV export and memory log show concrete actions and stateful behavior.